

Imaging Informatics: Long-tailed & Rare Condition

Team 2 Knowledge Inquiry

Teams Members

Shlok -



Andy -



Ismael -



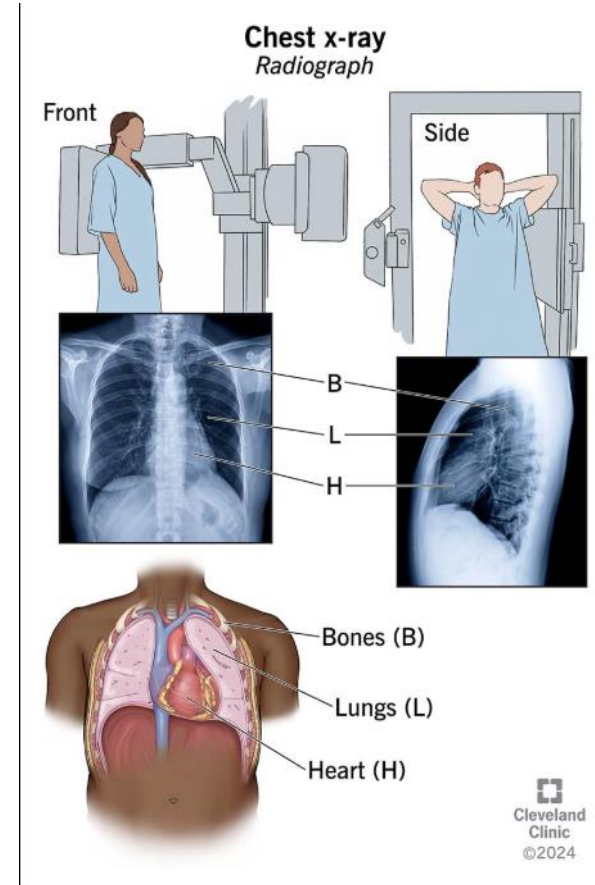
Sam -



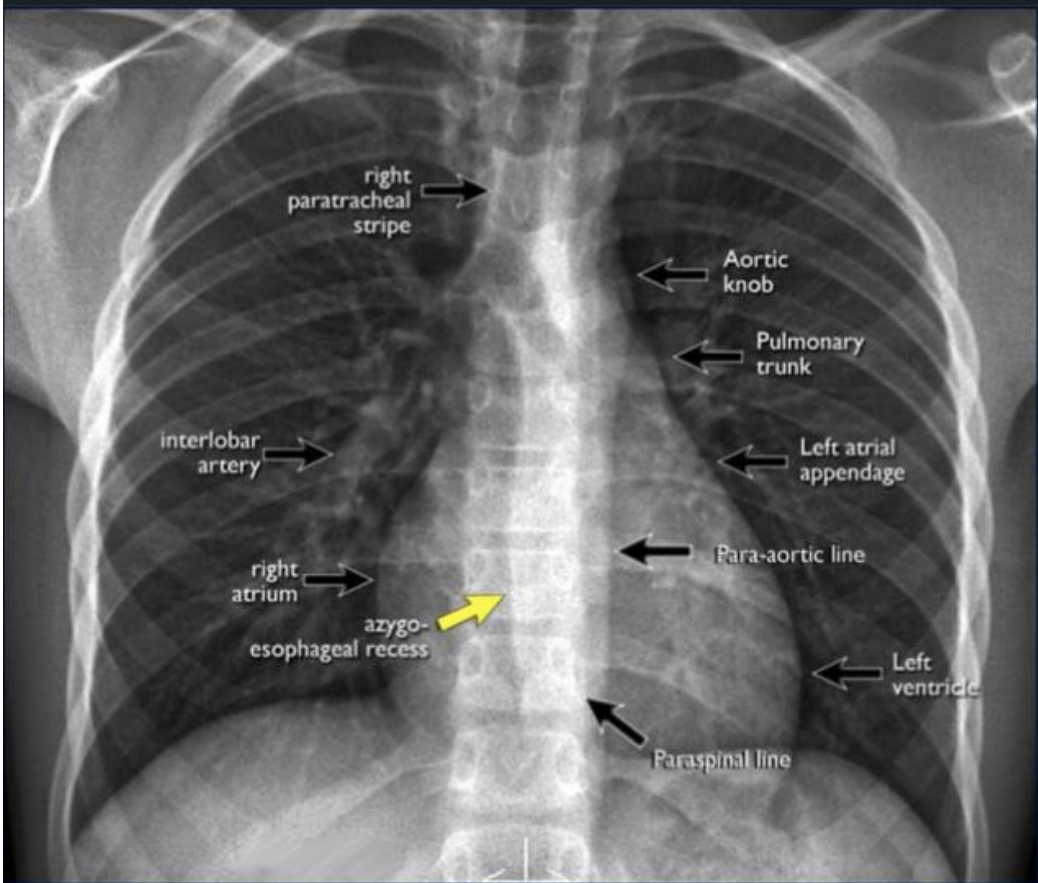
Introduction

Chest Radiography

- Chest Radiography also known as Chest X-ray (CXR) is a non-invasive medical imaging test that uses a small amount of ionizing radiation.
- Visualization -
 - Dense materials absorb more radiation and appear white.
 - Less dense materials, like air in lungs, allow radiation to pass and appear black.



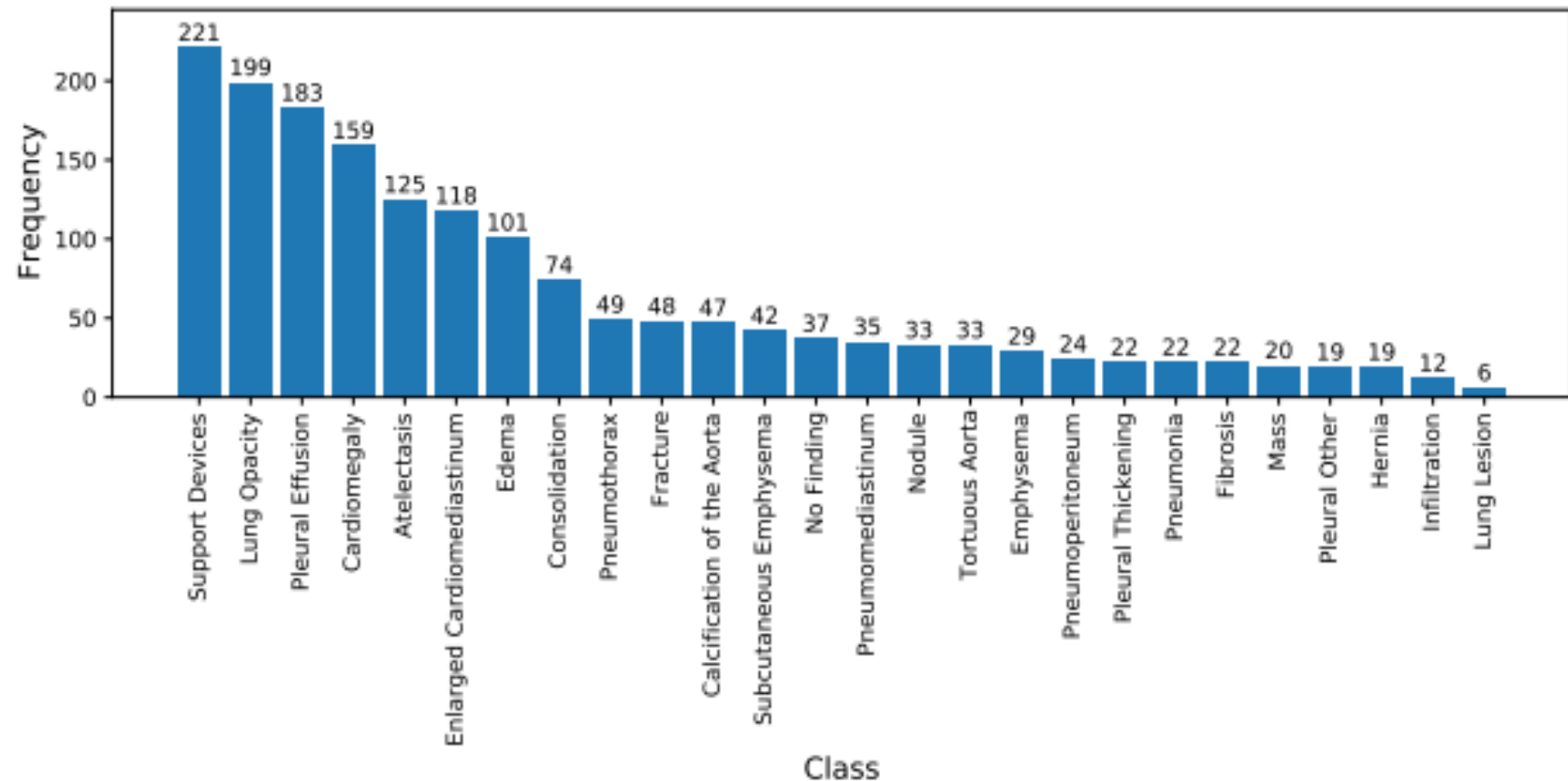
Anatomy



- CXR primarily used for lungs, heart, pleural space, skeleton, and mediastinum.
- Lung : pneumonia, fluid or tumors
- Heart : heart failure, cardiomegaly
- Pleural space: pneumothorax, fluid buildup

Long Tailed Nature of Radiology

- A small number of findings dominate the dataset
- Many clinically important findings occur rarely



Core Issue in Detection

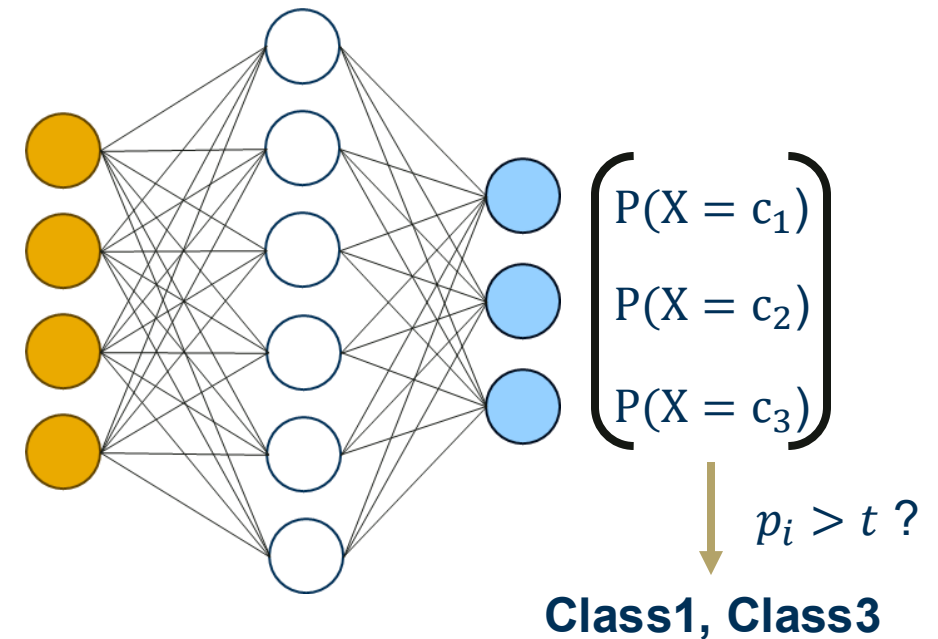
- Class imbalance skews loss toward frequent labels.
- Some rare conditions show extreme variability and large negative deltas

	Automated	Human	Δ
Atelectasis	0.488	0.308	-37%
Calcification of the Aorta	0.062	0.116	+88%
Cardiomegaly	0.448	0.392	-13%
Consolidation	0.214	0.182	-15%
Edema	0.313	0.249	-20%
Emphysema	0.113	0.071	-37%
Enlarged Cardiomediastinum	0.313	0.291	-7%
Fibrosis	0.081	0.054	-33%
Fracture	0.133	0.118	-11%
Hernia	0.074	0.047	-37%
Infiltration	0.113	0.03	-74%
Lung Lesion	0.076	0.015	-81%
Lung Opacity	0.571	0.49	-14%
Mass	0.096	0.049	-49%
No Finding	0.091	0.091	+0%
Nodule	0.096	0.081	-15%
Pleural Effusion	0.562	0.451	-20%
Pleural Other	0.052	0.047	-10%
Pleural Thickening	0.052	0.054	+5%
Pneumomediastinum	0.099	0.086	-12%
Pneumonia	0.283	0.054	-81%
Pneumoperitoneum	0.084	0.059	-29%
Pneumothorax	0.214	0.121	-44%
Subcutaneous Emphysema	0.108	0.103	-4%
Support Devices	0.564	0.544	-4%
Tortuous Aorta	0.059	0.081	+38%

Machine Learning with Long Tail Distributions

Core ML Concepts - Classification

- **Classification:** Models predict a label – **class** for each input
- **Multi-Label:** Inputs may have one or more classes
- **Predictions:** Models calculate a final vector of probabilities, one for each class – this is the probability the input is of that class
- **Decisions:** A label is chosen if the class probability is above a *threshold* (typically 0.5)



Core ML Concepts - Evaluation

Classifications (2 Classes + -)

- **True Positive (TP) / True Negative (TN)** : Model correctly predicts class of Sample
- **False Positive (FP)**: Model predicts positive (+), but sample was negative (−)
- **False Negative (FN)**: Model predicts negative (−) but sample was positive (+)

Classifications (N Classes)

- For each class j , set j as the positive and all others as negative
- Compute TP/TN/FP/FN for all classes

Metric	Definition	Formula
Accuracy	Overall Correctness	$\frac{TP + TN}{TP + TN + FP + FN}$
Precision	Accuracy of positive predictions	$\frac{TP}{TP + FP}$
Recall	Ability to find positive instances	$\frac{TP}{TP + FN}$

Chest X-Ray Classification Challenges

Key DL Challenges: **Long-Tailed** and **Multi-Label** Classification

Long-Tailed Classification

- Traditional ML models struggle with classification over long tailed distributions
- Models exhibit bias to the most common classes ignoring the critical rare condition

Multi-Label Classification

- Patients often demonstrate multiple conditions at once
- Traditional training techniques rarely incorporate '*label co-occurrence*' into training

Chest X-Ray Evaluation Challenges

Mean Area Under the ROC Curve (mAUC)

- AUC measures the probability that a model will rank a positive sample higher than a negative sample
- It is insufficient for long-tailed datasets as the false-positive rate is low due to the rare classes

$$Pr = \frac{TP}{TP + FP} \quad Re = \frac{TP}{TP + FN}$$

$$TPR = \frac{TP}{TP + FN}, \quad FPR = \frac{FP}{FP + TN}$$
$$mAUC = \frac{1}{C} \sum_{i=1}^C \int_0^1 TPR_i(FPR_i) dFPR_i$$

Mean F1 Score

- The F1-score is the harmonic mean of Precision and Recall
- A classification is considered if it has a probability above a threshold
- F1-Scores are highly sensitive to decision thresholds

$$F1 = 2 \frac{Pr \cdot Re}{Pr + Re}$$

$$mF1 = \frac{1}{C} \sum_{i=1}^C F1_i$$

Mean Average Precision (mAP)

- Evaluates the Precision-Recall curve with a weighted mean of precisions at each threshold.
- mAP only accounts for the "positive" class. In imbalanced datasets, it penalizes models that produce many False Positives

$$AP = \sum_n (R_n - R_{n-1}) \cdot P_n$$

$$mAP = \frac{1}{C} \sum_{i=1}^C AP_i$$

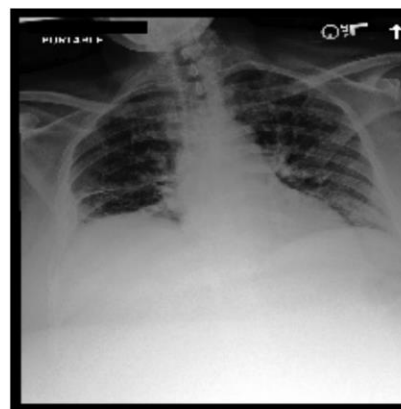
CXR and Long Tail Dataset

The CXR-LT Dataset

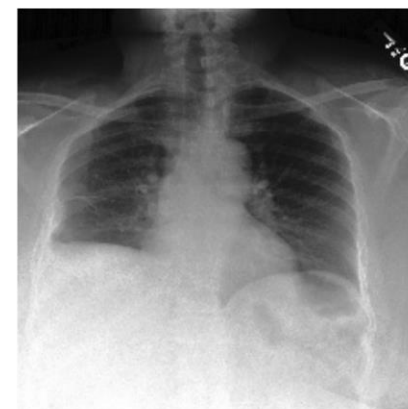
CXR-LT is a dataset/benchmark presented at ICCV 2023/MICCAI24 to promote developing new models specialized in long-tailed and multi-label classification of CXRs

Summary

- Given a CXR detect ***all*** findings
- **337,110** CXRs with *at least* 1 label
- **45** total clinical findings (classes)
- Extends the MIMIC-CXR-JPG dataset



(a) Hilum, Cardiome-diastinum, Fissure, Nodule, Pleural Effusion



(b) Fracture, Pleural Effusion



(c) Cardiomegaly, Edema, Lung Opacity

Example CXR Images and Labels in CXR-LT Dataset

- | | | | | | | | |
|------------------------------|--------------------------|-------------------------------|--------------------|---------------------|----------------------|--------------------------|--------------|
| • Calcification of the Aorta | • Nodule | • Tortuous Aorta | • Fracture | • Pleural Other | • Hydropneumothorax | • Pulmonary Hypertension | • Hilum |
| • Emphysema | • Pleural Thickening | • Atelectasis | • Lung Lesion | • Support Devices | • Infarction | • Rib Fracture | • Osteopenia |
| • Fibrosis | • Pneumomediastinum | • Cardiomegaly | • Lung Opacity | • Adenopathy | • Kyphosis | • Round Atelectasis | • Scoliosis |
| • Hernia | • Pneumoperitoneum | • Consolidation | • Pleural Effusion | • Azygos Lobe | • Lobar Atelectasis | • Tuberculosis | • Normal |
| • Infiltration | • Subcutaneous Emphysema | • Edema | • Pneumonia | • Clavicle Fracture | • Pleural Other | • Bulla | |
| • Mass | | • Enlarged Cardiome-diastinum | • Pneumothorax | • Fissure | • Pulmonary Embolism | • Cardiomyopathy | |

CXR-LT Evaluation

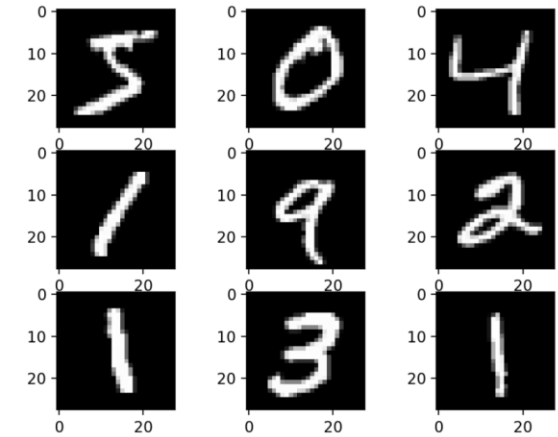
	Threshold Independent?	Sensitive to Imbalance?	Primary Focus
mAUC	Yes	High	Ranking/Separability
mF1	No	Moderate	Specific Decision Point
mAP	Yes	Low	Precision-Recall trade-off

- **Resilience to Imbalance:** mAP ignores True Negatives, preventing the metric from being inflated by a model that simply predicts "no disease" for everything – suitable to long tailed datasets
- **Threshold-Agnostic:** mAP evaluates models independent of a decision threshold
- **Rare Classes:** By using a **macro-average** (averaging the 26 classes equally), the score is not inflated by success on the "no disease" class

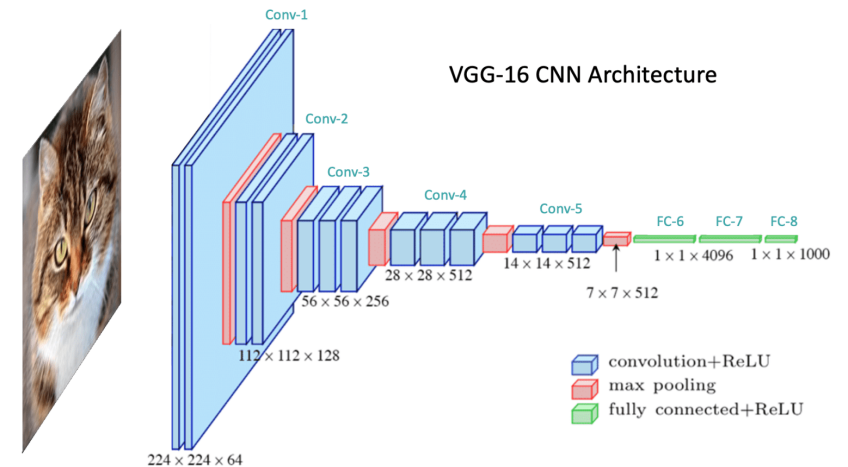
Timeline of ML Models From Vision to Language

Vision Models

- Vision Models simply are Machine Learning Models that allow image processing
- The idea has been around for a while
 - Most famously CNNs for MNist Hand-Writing Classification released in 1999
- There are many Vision Models around, most famously are CNNs (Convolution neural networks)
 - Allows for classification of images, introduced by Bell Labs in 1989
- Another famous technique is ViTs (Vision Transformers), which operate similarly to LLMs
 - Breaking down an image into tokens



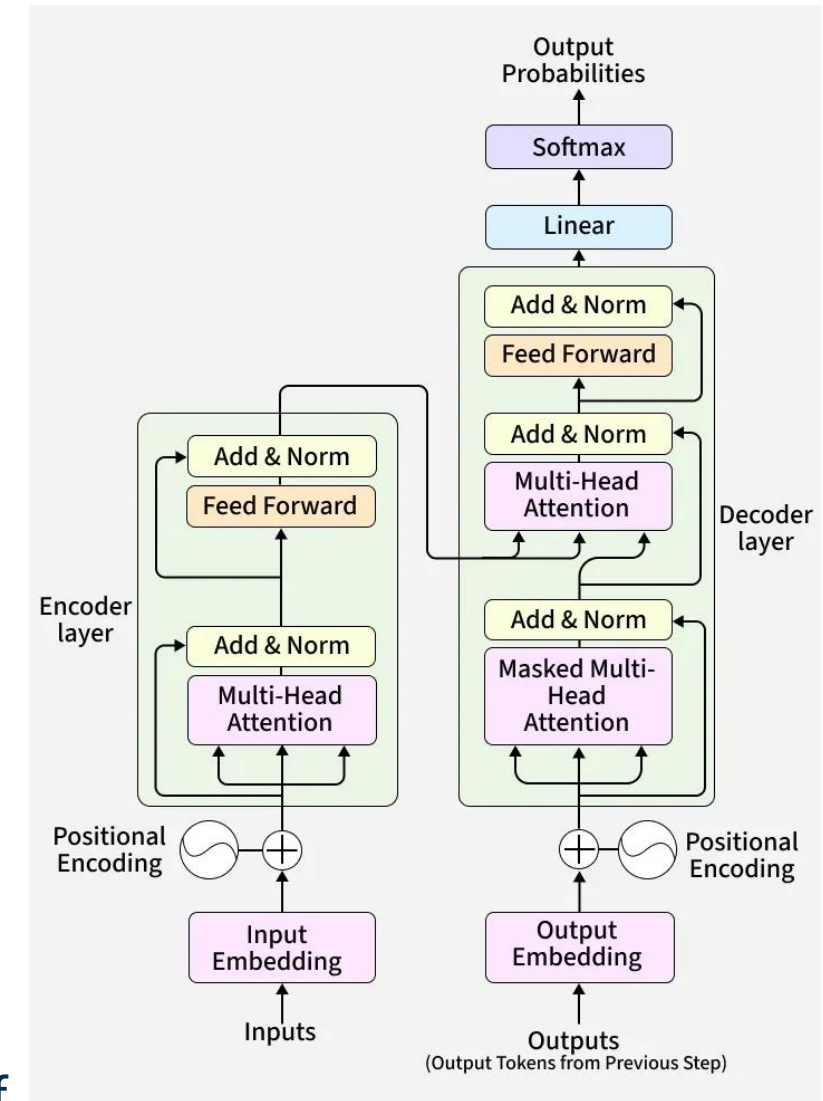
MNist Data Sample



Sample Layout of a CNN

Transformers

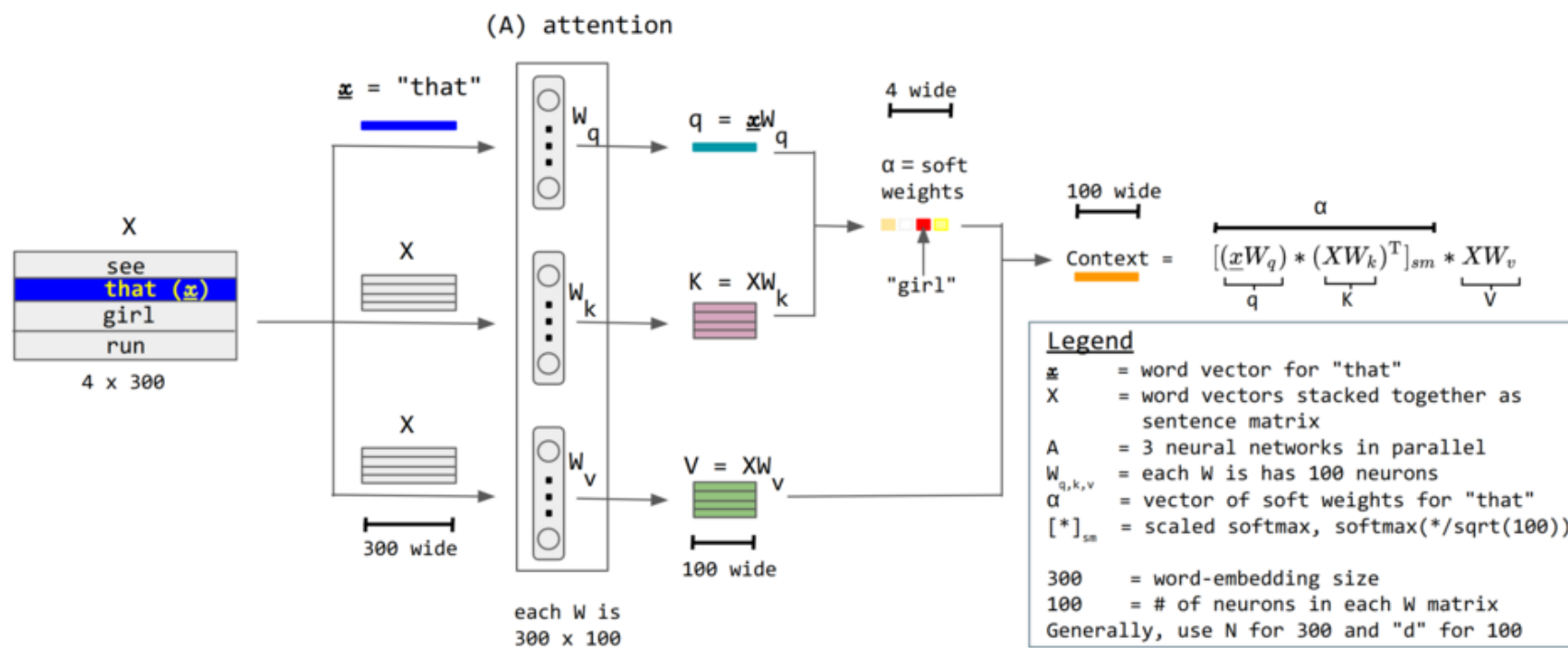
- A deep learning architecture based entirely on the **Attention Mechanism**, discarding recurrence and convolutions.
 - "Attention is all you need" published by Google in 2017
- Transformers typically have 2 "layers"
 - Encoder: Processes input into a vector for model to process
 - Decoder: Translates Vector to Output
- The Attention Mechanism is fancy math that ranks the importance of each part of the input to other parts
 - Unlike Convolution or fully connected layers, where all parts of input have an equal contribution. Attention allows model to focus on only important parts



Transformer Architecture

Attention Layer

- Takes input X , and breaks into query (q), Key (K), and Value (V)
 - q is the word/pixel/token of interest, K and V are trained weights that allow the model to correlate the importance of words against each other
- Output is a matrix showing relative importance of each q to rest of input



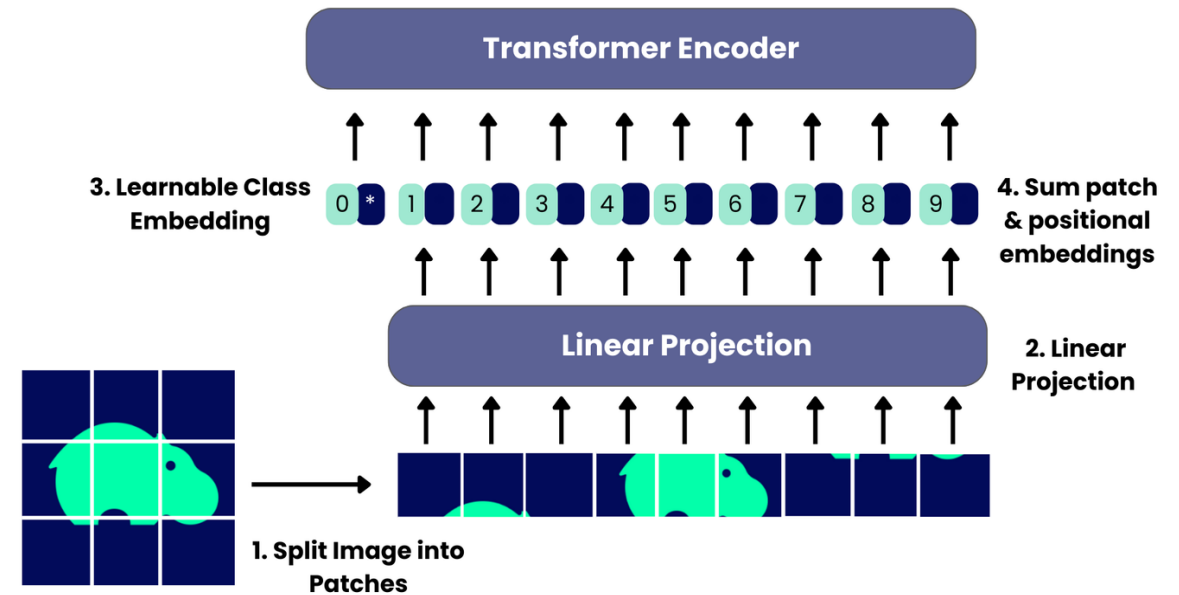
	I	love	you
je	0.94	0.02	0.04
t'	0.11	0.01	0.88
aime	0.03	0.95	0.02

Example of Attention Output for translation

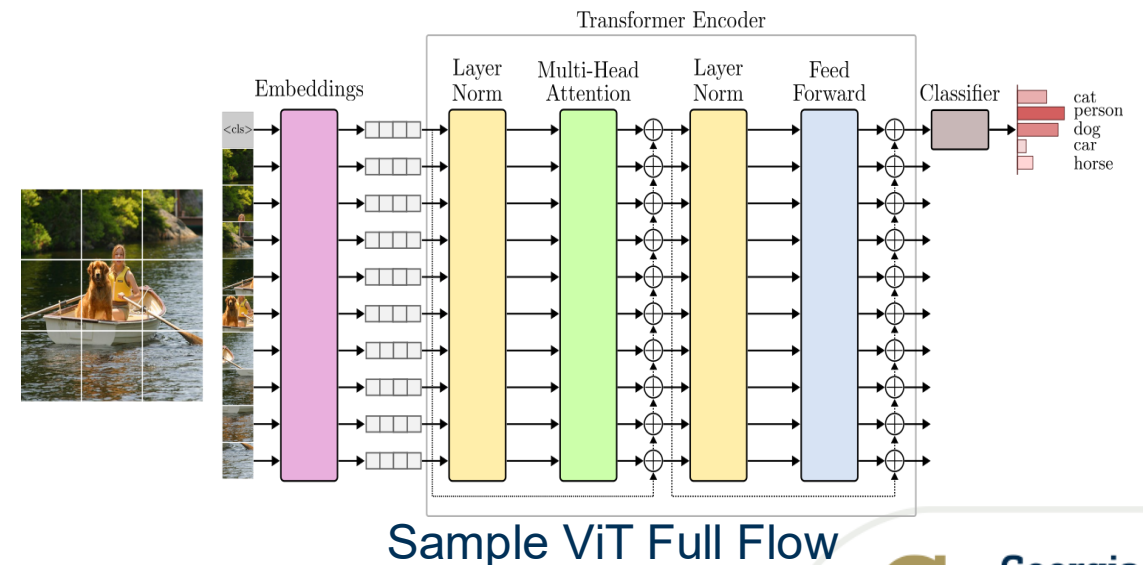
Walkthrough Example of Attention Layer

Vision Transformers

- Compared to CNNs, ViTs are much flexible and generalizable at image classification introduced by Google in 2019
- ViTs take an image (of any size) and cut its pixels up into patches (ex: 16x16)
- Each Patch is projected to a vector embedding which can then be used to train a model
- CNNs have a fixed input and a fixed set of outputs, making them difficult for generalization
- ViTs also naturally interface with LLMs



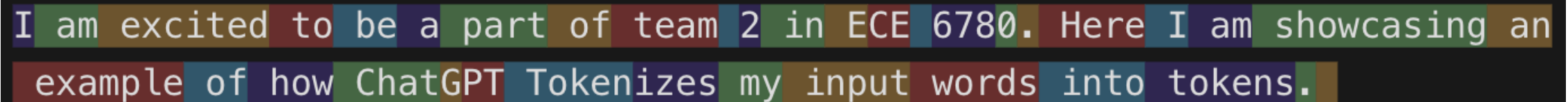
ViT Encoding Workflow



Sample ViT Full Flow

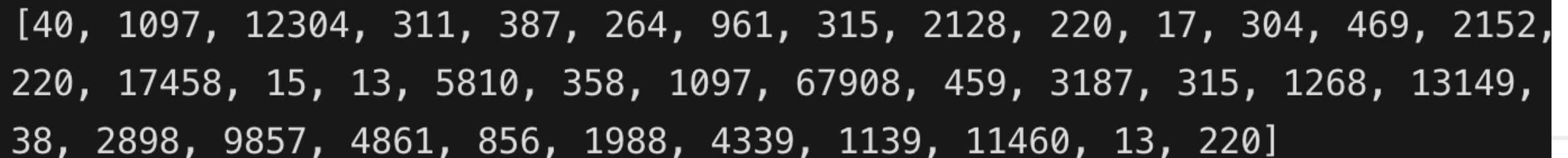
Large Language Models (LLMs)

- LLMs (like ChatGPT, Claude, Gemini) all operate using transformers introduced by OpenAI in 2018
- They convert your input text into "tokens" which are inputs to the model
 - A token is a chunk of text that is translated into a number.
 - For ViTs, a token is a group of pixels instead of text



I am excited to be a part of team 2 in ECE 6780. Here I am showcasing an example of how ChatGPT Tokenizes my input words into tokens.

Tokens of a sentence by ChatGPT-4

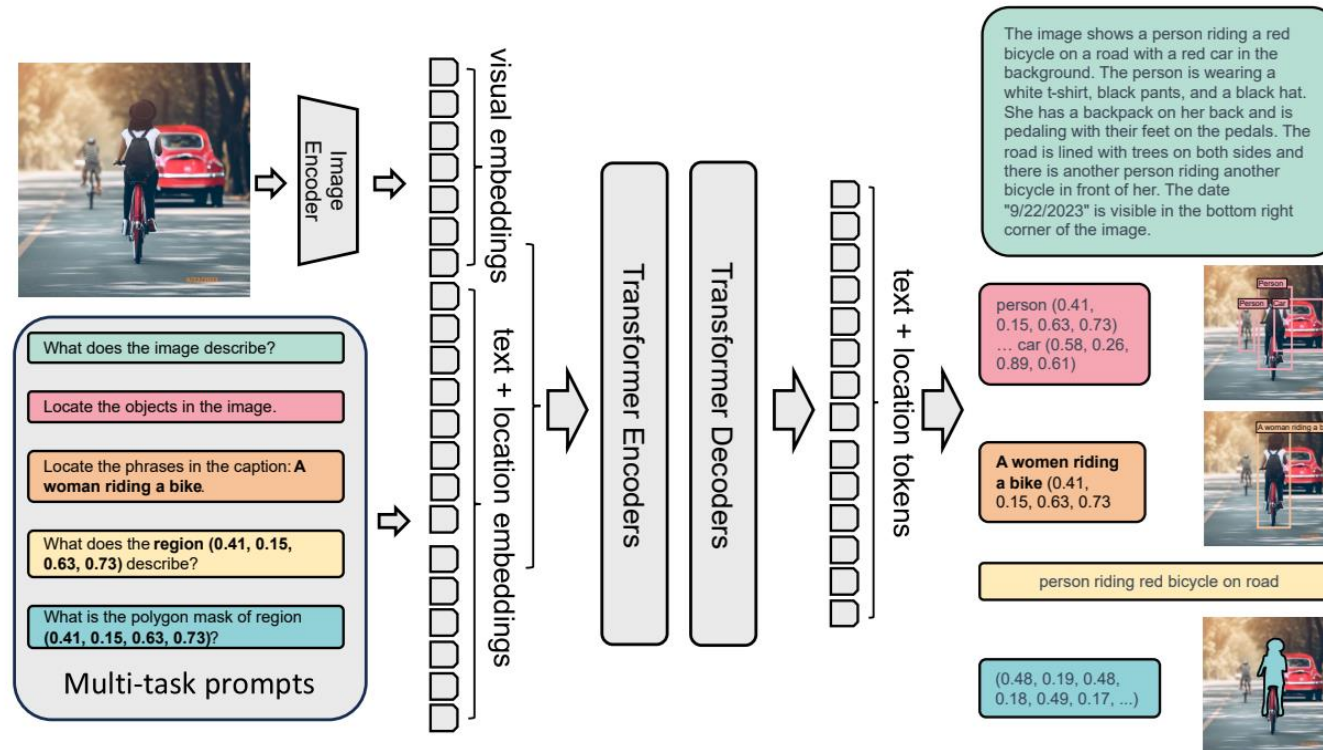


[40, 1097, 12304, 311, 387, 264, 961, 315, 2128, 220, 17, 304, 469, 2152, 220, 17458, 15, 13, 5810, 358, 1097, 67908, 459, 3187, 315, 1268, 13149, 38, 2898, 9857, 4861, 856, 1988, 4339, 1139, 11460, 13, 220]

Token IDs for above tokenized sentence

Vision Language Models (VLMs)

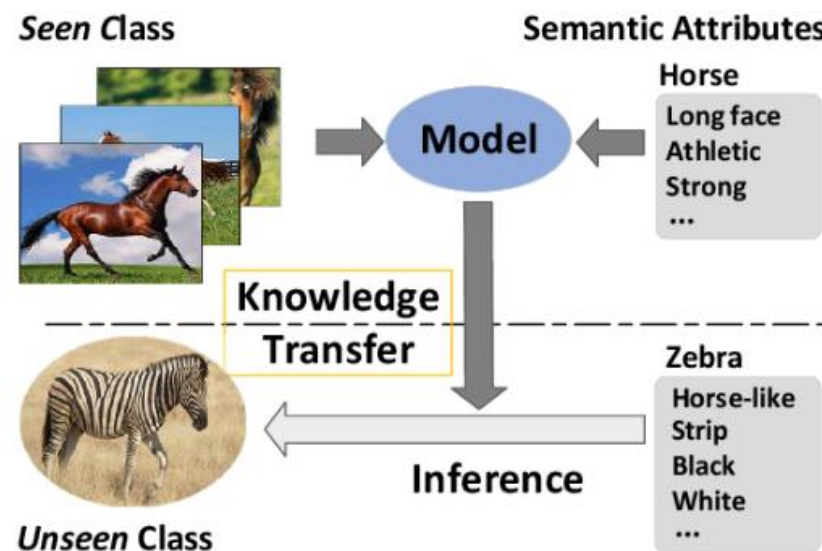
- VLMs are a "hybrid" between ViTs and LLMs, allowing the co-processing of images and text, first done by Microsoft, UW, and Columbia in 2023
- Image and text are both encoded to input "tokens"
 - Text through regular tokenization, image through an image encoder



Example of VLM flow

Zero Shot Learning

- Zero-Shot learning is an ML technique where a model is tested on scenarios it has never seen before, can it transfer its knowledge to learn?
 - Concept pre-dates LLMs, idea introduced in 2008 by University of Montreal for Natural Language Processing
- For our project, the goal of zero-shot learning is to assess whether the model can correctly predict that a given X-ray is for a disease that it hasn't been seen before
 - A nice bonus would be if the model can predict what that unseen disease is



Current Literature

CXR-CML: Improved zero-shot classification of long-tailed multi-label diseases in Chest X-Rays

Context and Objective

- Long-tailed disease distribution (similar to CXR-LT)
- Standard CLIP struggles with rare classes
 - Contrastive Language–Image Pretraining (VLM)

Goal: Improve zero-shot classification for tail diseases

Approach: Restructure embedding space to better separate rare classes

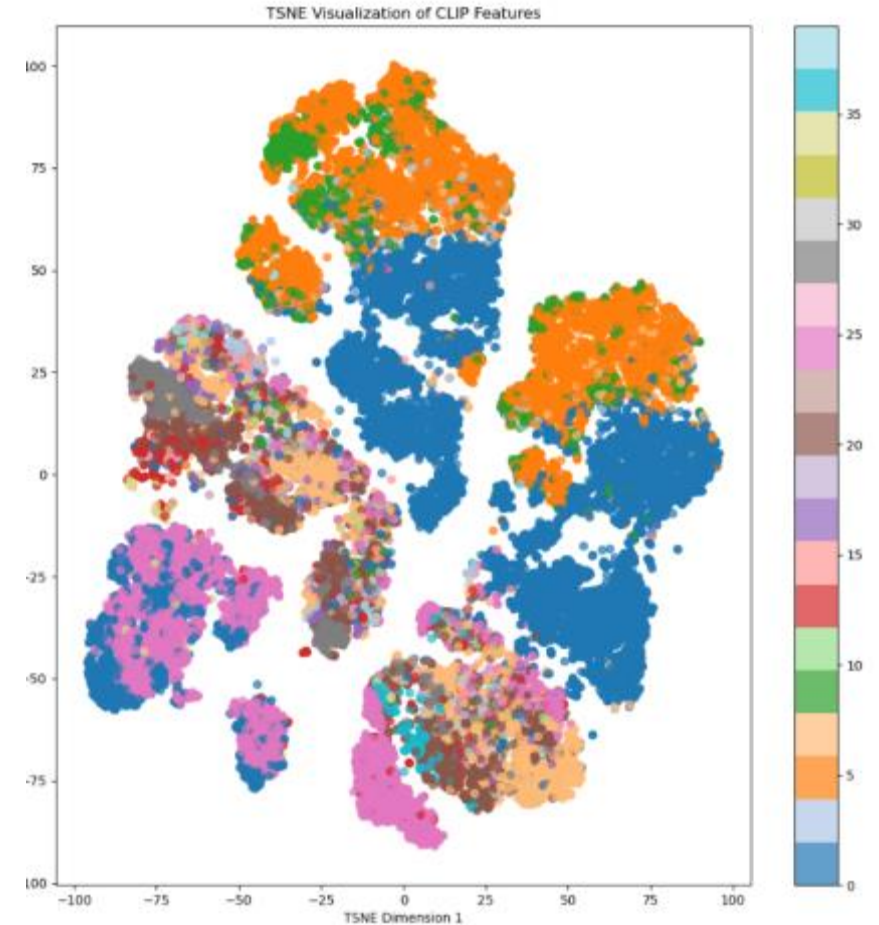
Table 1: Distribution of disease classes based on the number of samples.

Class Category	Number of Classes
Common ($>10,000$ samples)	11
Medium (1,000–10,000 samples)	17
Rare ($<1,000$ samples)	12

Problem with Standard VLMs

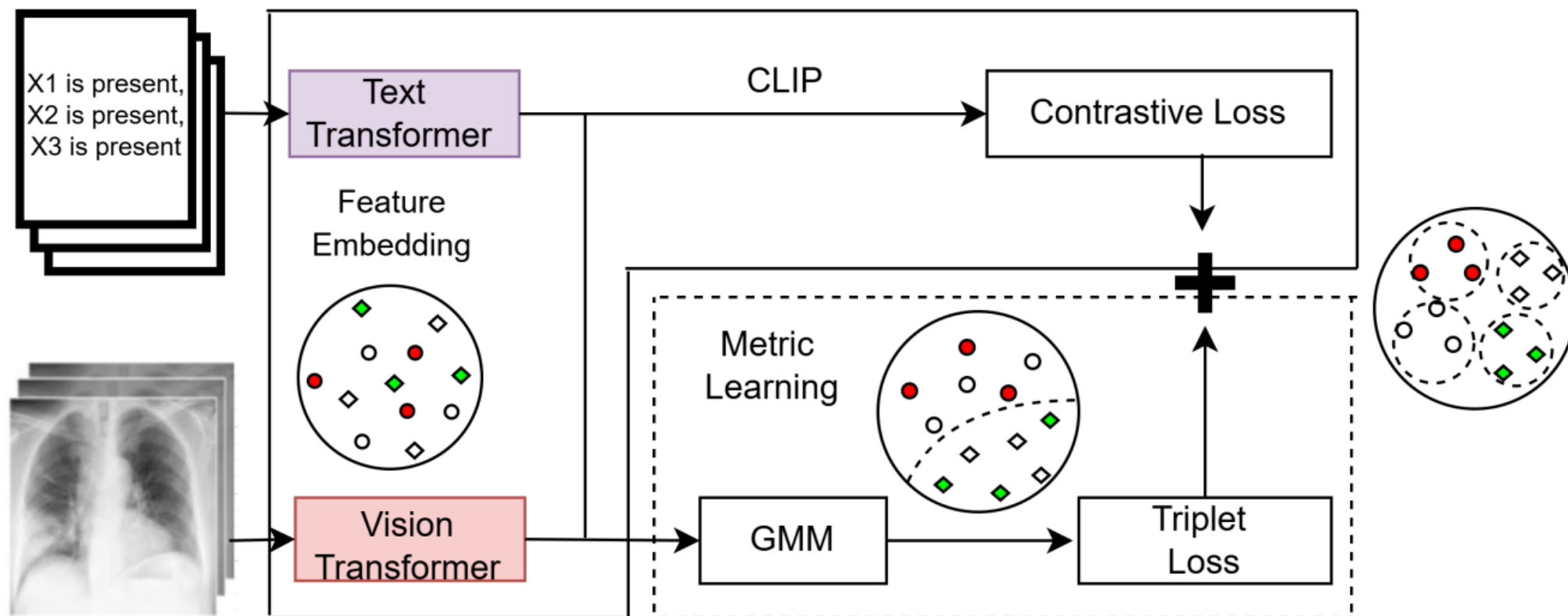
Plot

- Colors represent different diseases
- Significant overlap between disease clusters
- Alignment objective treats all classes equally



(a) Latent space of CheXzero

Methods



Results

- Rare AUC improves
 - Improves mAP in CXR-LT
- No impact on base classes
- Implied Improvement in Zero Shot

Method	Total AUC	Base AUC	Rare AUC
CheXzero (Baseline) [7]	0.644*	0.647*	0.631*
CheXzero + Meta labels	0.691*	0.681*	0.707*
CXR-CML ($bs = 32, \nu = 2$)	0.710	0.710	0.710
CXR-CML ($bs = 32, \nu = 4$)	0.715*	0.711*	0.720*
CXR-CML ($bs = 32, \nu = 6$)	0.715	0.710	0.718
CXR-CML ($bs = 16, \nu = 4$)	0.705	0.700	0.710
CXR-CML ($bs = 32, \nu = 4$)	0.715*	0.711*	0.720*
CXR-CML ($bs = 64, \nu = 4$)	0.711	0.707	0.716

Takeaway:

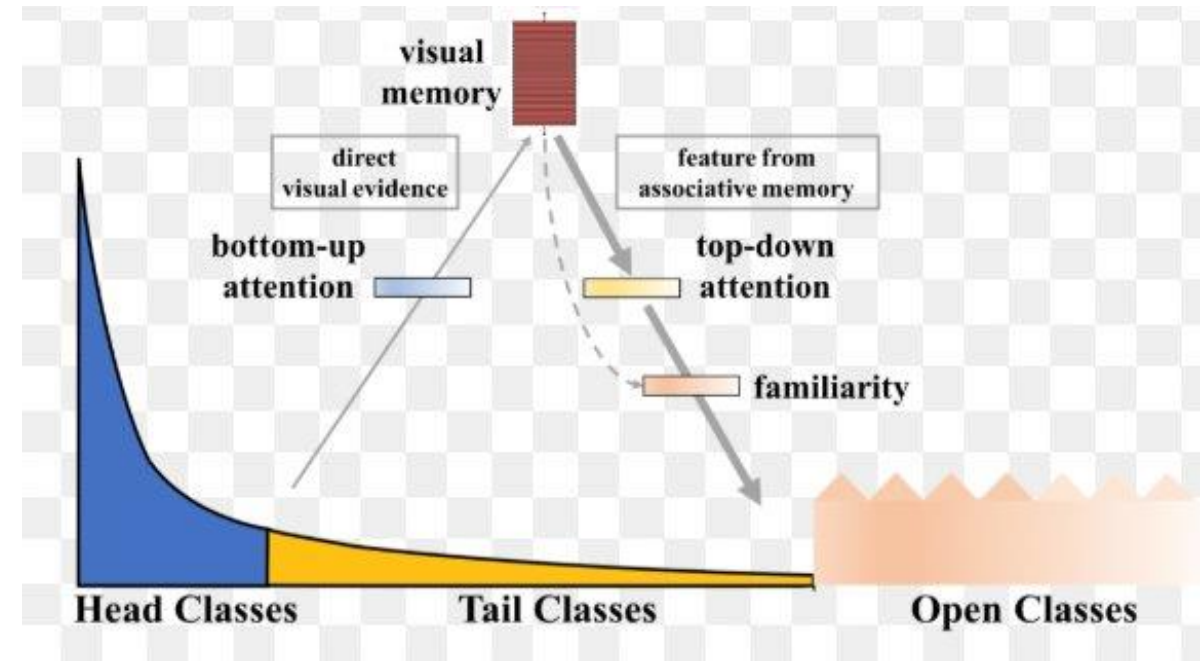
Embedding structure must account for imbalance to improve rare-class and zero-shot robustness.

Literature Review #1 -

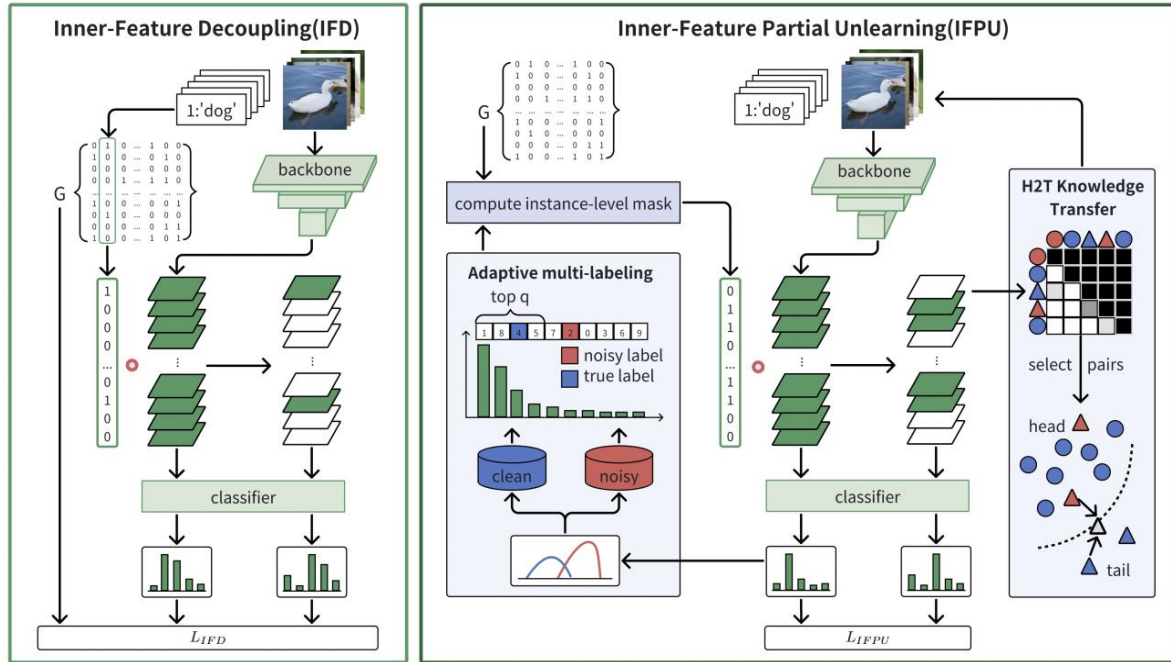
Classifying Long-tailed and Label-noise Data via Disentangling and Unlearning

Objective

- Improve model classification on datasets where rare "tail" classes are frequently mislabeled as common "head" classes.
 - Long tailed distributions lead to increased noisy labels
 - Tail class associated with head features
 - Robustness of learning



Methods



- DULL (Disentangling and Unlearning for Long-tailed and label-noisy data)
 - Inner Feature Disentangling
 - Inner Feature Partial Unlearning
 - Adaptive multi – labeling and knowledge transfer

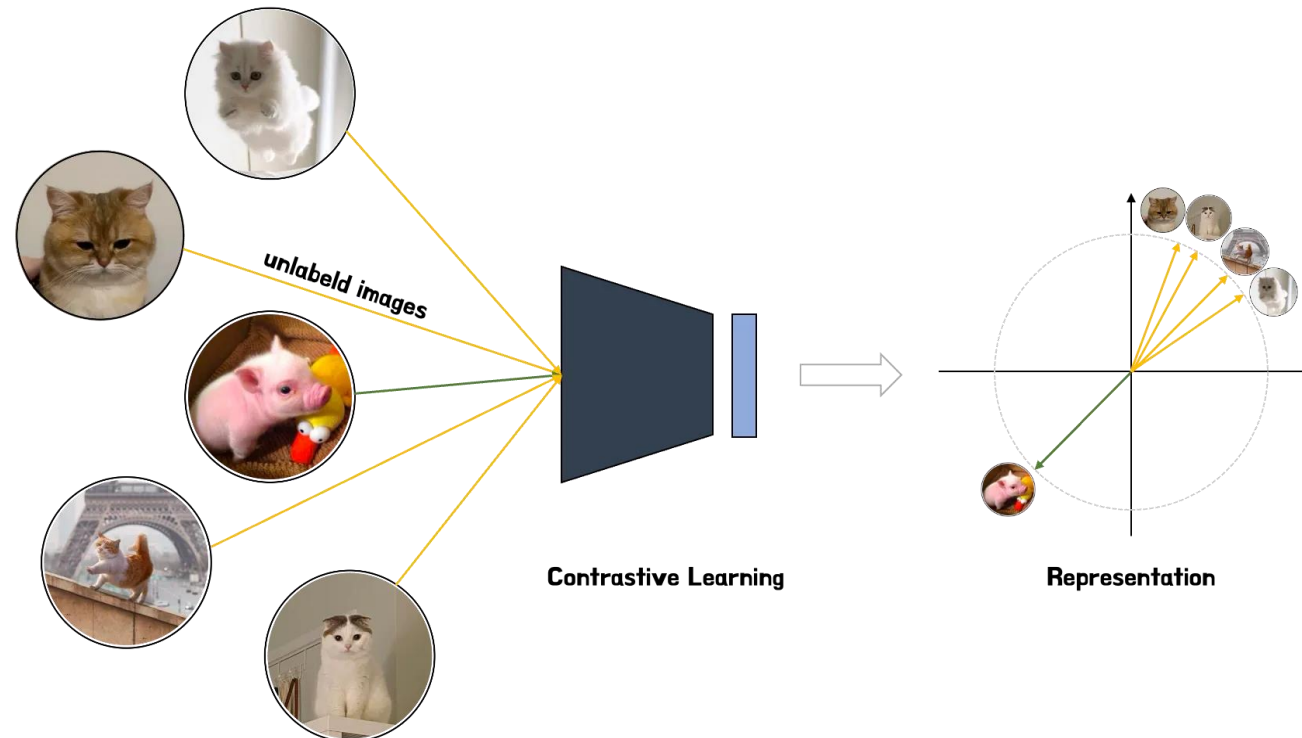
Results

- Evaluated DULL against various baselines on simulated and real-world datasets
 - Superior accuracy: DULL consistently outperformed existing long-tail (LT) and noise (NL) learning methods
 - On CIFAR-100 with 40% T2H noise, DULL achieved 46.48% accuracy, compared to 34.64% for the baseline
 - In real-world T2H scenarios, DULL improved performance by 6.73% over previous methods
- DULL showed significant improvements across head, middle, and tail classes rather than improving one at the expense of other

Specific VLMs and LT techniques

How to correlate images to text?

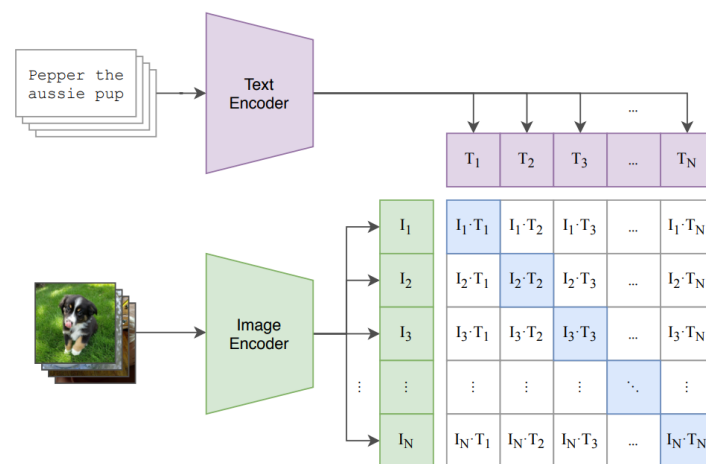
- **Contrastive learning** is a method of self-supervised learning where the model is fed few labelled images and the model predicts how similar or different unlabelled images are to the labeled.
 - The result is a space where similar images are close together and different images are far apart
 - In the case of our project, the labelled image data is CXR, and the ideal resultant space would have similar conditions near each other, and far from other conditions.



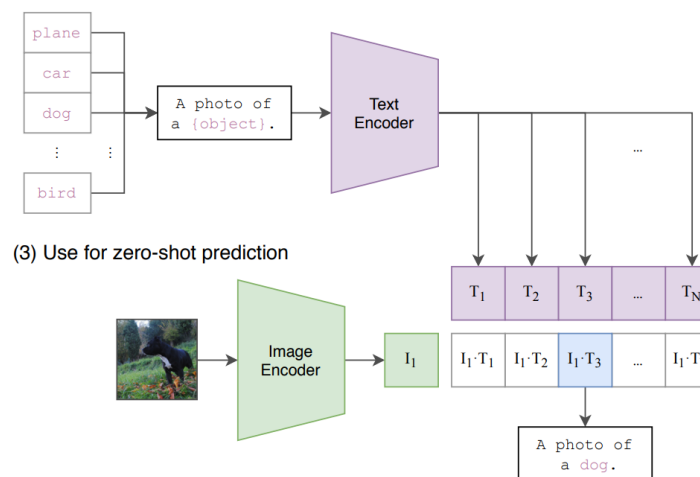
Contrastive Learning

- To train the model, the labelled data can be used
 - An "Anchor" image is used for each condition, and the rest of the data is labelled "positive" if it has the same condition, "negative" otherwise.
- There are many techniques to then train the model on similarity, but most scalable and best-performing is a technique called "SigLIP" by Google in 2023
 - Encode each text and image pair into a image and text vector (tokenization/image encoder)
 - Create a similarity matrix b/w text and images (dot-product b/w text and image vector)
 - Create labelled matrix for whole matrix (each image has a 1 for its caption, 0 otherwise)
 - Finally do sigmoid loss function to compute similarity (high value = match, low = no match)

(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

Retrieval-Augmented Generation

Generative models can "**hallucinate**" when lacking specific knowledge or context

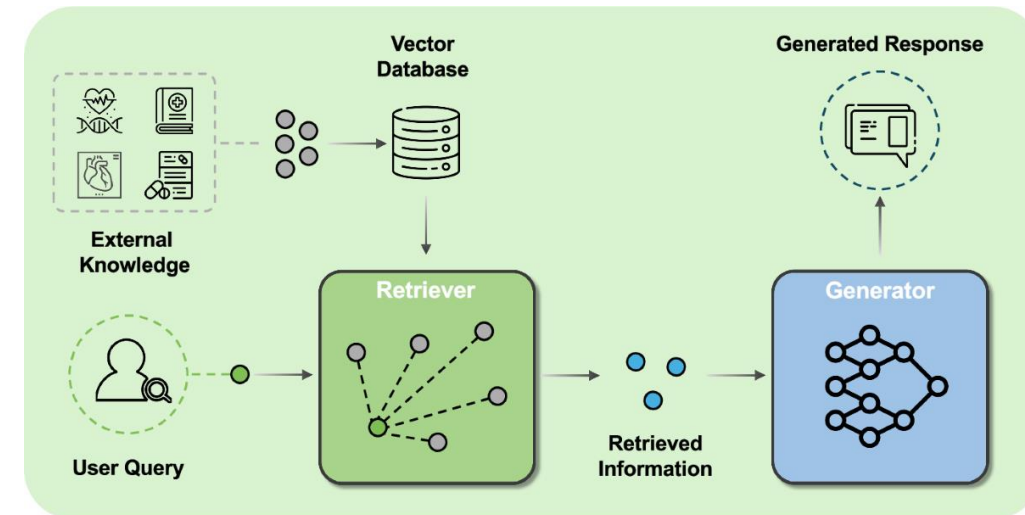
Solution: RAG - Instead of relying on internal parameters, models retrieve relevant external data to construct an answer.

Multi-Modal RAG for VLMs - CXR

- Retrieve **Text**: Clinical guidelines, past diagnostic reasoning trees, or similar patient notes
- Retrieve **Images**: Retrieving similar historical X-rays to compare with the current case.

Process

- **Query**: Ask VLM: "Does this patient have a nodule?"
- **Retrieve**: Fetch similar confirmed nodule cases from a database
- **Generate**: Produce an answer conditioned on both the current image and the retrieved evidence



Data Re-Balancing

Solution: Resampling – Modify the training data distribution to prevent bias toward majority classes

Oversampling (ROS)

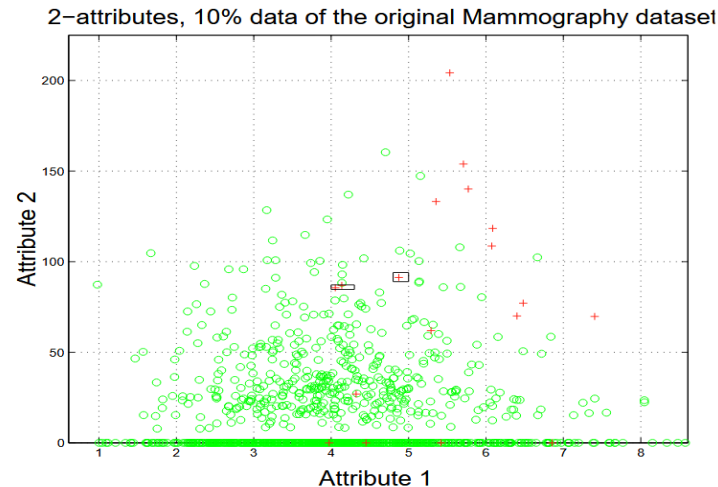
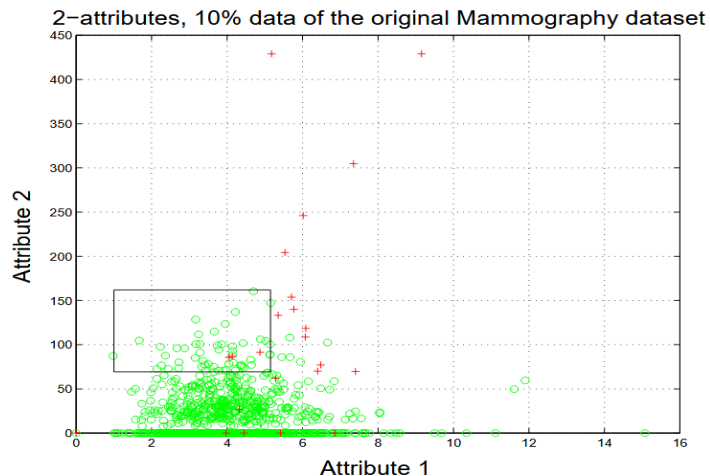
- Randomly duplicate + augment images from rare classes increasing frequency
- May lead to overfitting, memorizing duplicates rather than learning class features

Undersampling (ROS)

- Sample less point from majority class to match frequency of tails
- May lead to underfitting and a weaker feature extractor

Class-Balanced Sampling

- Construct training batches to contain equal numbers of samples from each class



Decision-Tree classifier before and after over sampling with minority classes

Cost-Sensitive Learning

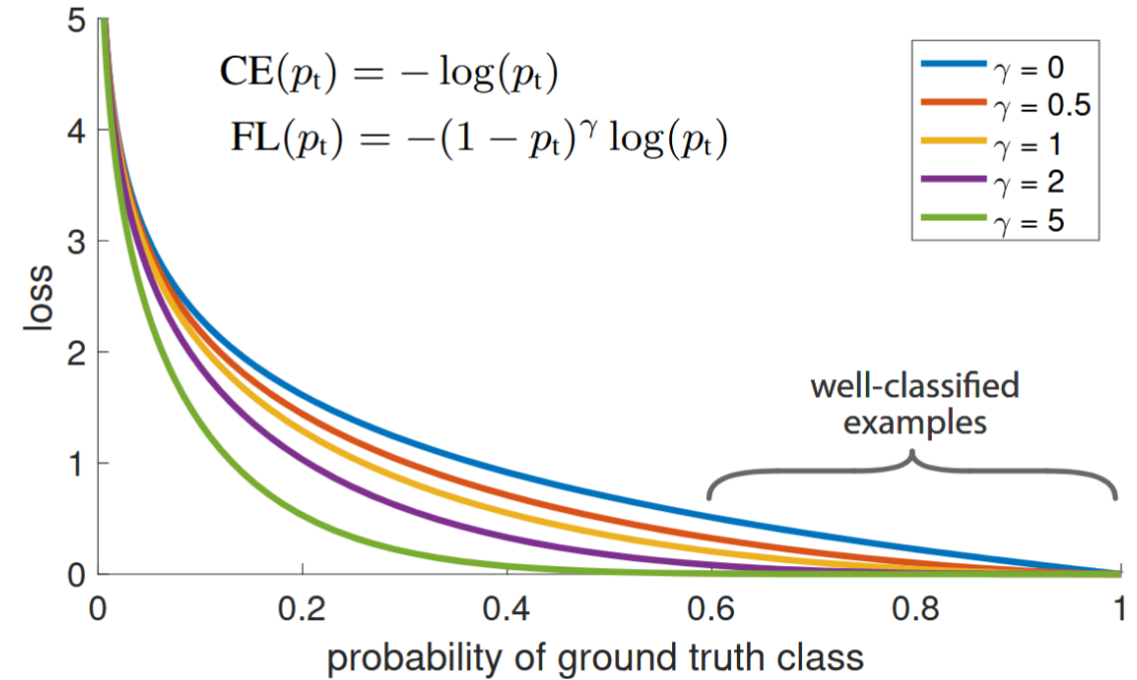
Solution: Reweighting – Increase loss contributed by rare classes

Weighted Cross-Entropy Loss

- Assign a weight to the loss for class, inversely proportional to class size
- $w_c \propto 1/N_c$
- Misclassifying a rare disease is penalized with a large loss, steering the optimizer

Focal Loss

- Dynamically scale loss based on how “hard” an instance is to classify
- $FL = -\alpha(1 - p_t)^\gamma \log(p_t)$ - (p_t is the probability of the class)
- As p_t approaches 1 (high-confidence) the loss goes to 0 limiting focus on easy classes (Head Classes)



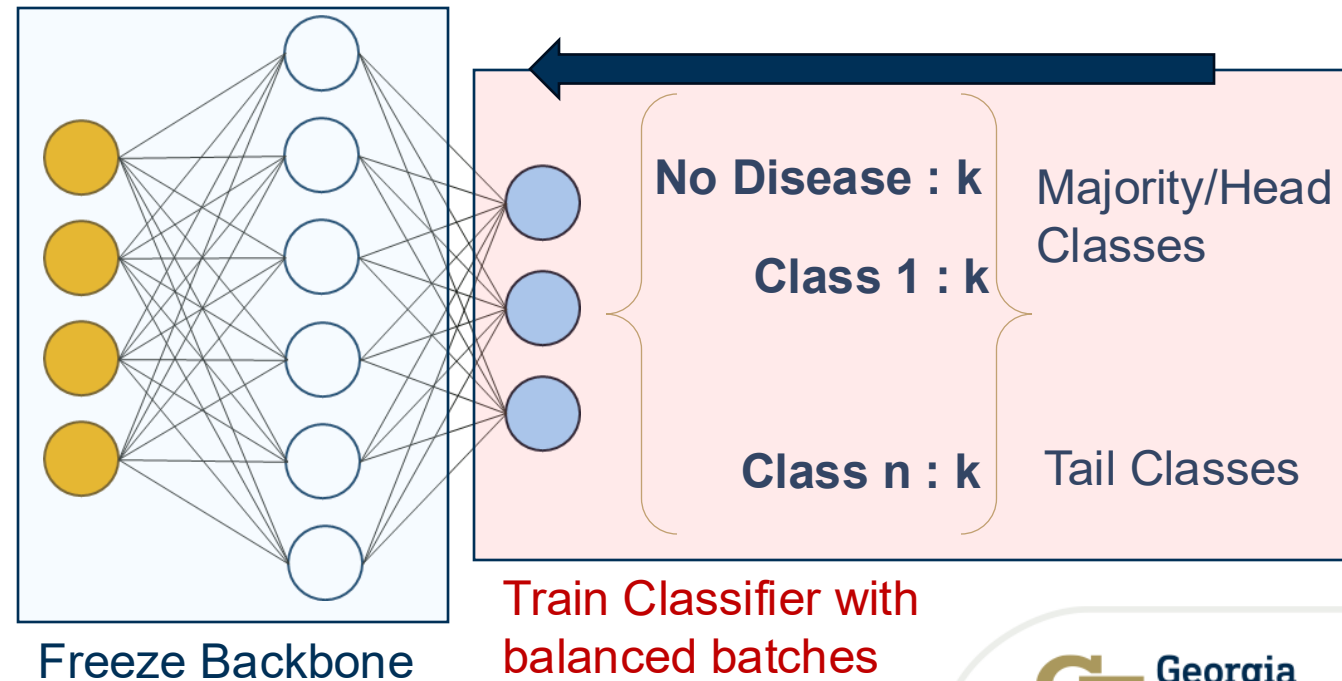
Decoupled Training

Solution: Separate feature extraction learning and decision making learning

- Feature extractors may learn good features, but classification layers bias towards majority classes
- Separate into **Representation Learning** and **Classifier Balancing**

Classifier Balancing

- Freeze the feature extractor
- Re-train the classification layer with **Class-Balanced Sampler**
- Fair decision making without biasing feature extraction



Head-to-Tail Knowledge Transfer

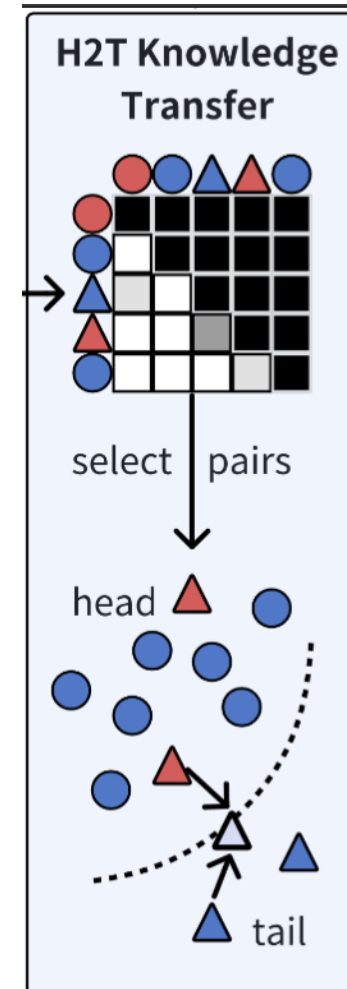
Solution: Synthesize new samples for rare classes with feature rich data from majority

Semantic Sample Selection

- **Similarity-Based:** Pair head and tail instances that are visually similar (inner product of masked features)
- **Directional-Constraints:** Zero out generations which supplement head instances with tail features

Mix-Up

- **Blend:** Compute a 50/50 blend of the tail and head images, with a blended label (LERP)
- Artificial images are added to training set increasing volume of rare disease cases with robust 'head' features



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Appendix

GLoRIA: A Multimodal Global-Local Representation Learning Framework for Label-efficient Medical Image Recognition

Problem and Gap

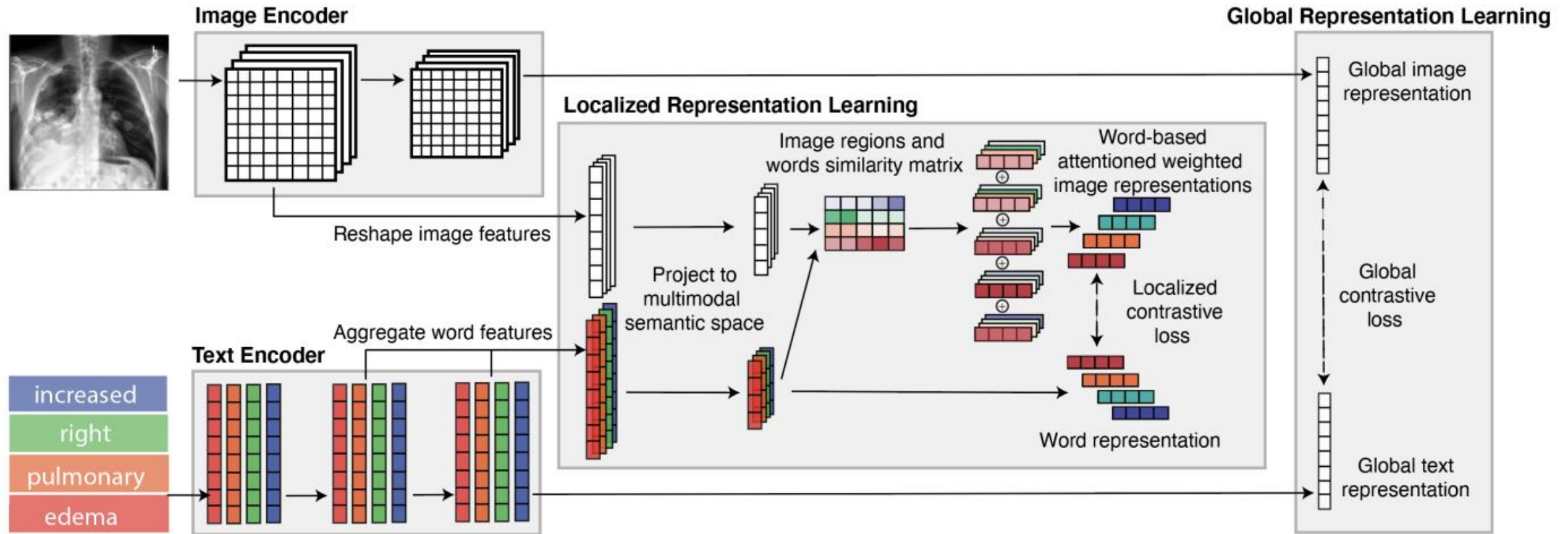
Previous Works:

- Most medical image models are trained with structured labels only
- Radiology reports contain underutilized semantic detail
- Models require a manually labeled dataset
- Multimodal approaches focus on global image–text alignment

Gap:

- Global alignment ignores region–word correspondences
- Limited support for label-efficient learning
- Poor interpretability for subtle or rare abnormalities

Methods



Findings

- Multimodal Training improves representation
 - Outperforms visual-only
- Supports zero-shot performance
 - With image-text embedding space
- Enables word to region attention alignment

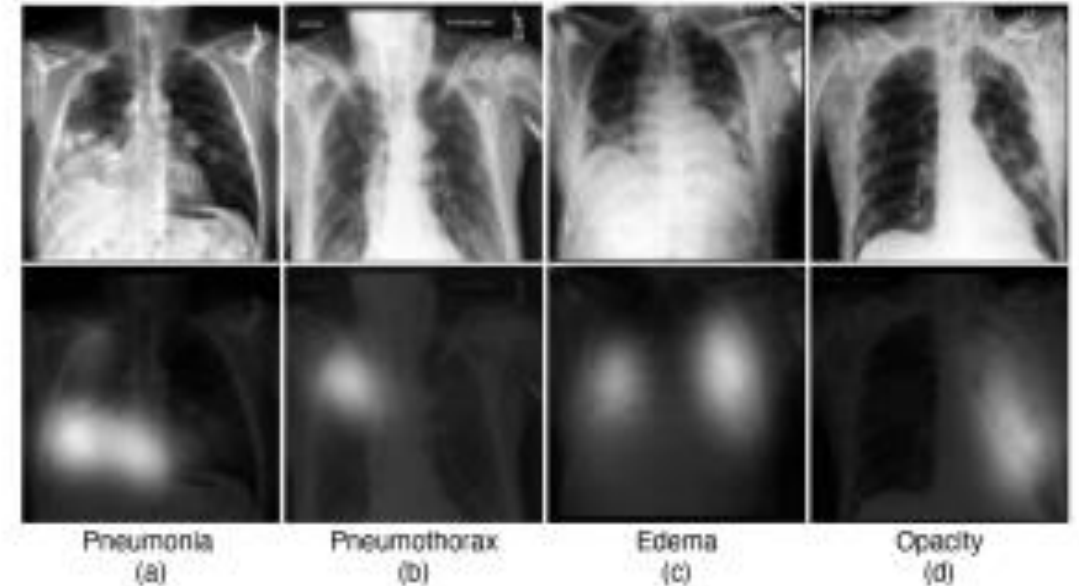


Figure 4: Examples of frontal radiographs of the chest (top) with corresponding attention weights for the given word (below).